

From Three Messy Sources to One Clean Pipeline

Week 3 — Multi-Source Operational Data Pipeline
Fuel Processing Plant · Nairobi, Kenya · June 2026

The Problem

Operational decisions at the plant rely on data sitting in three completely separate places — none of which talk to each other:

- **Internal sensors** log pressure, temperature, and flow rate every 10 minutes to a CSV file
- **External weather** data lives behind a public API — never queried, never connected to ops
- **Holiday & shutdown schedules** sit in a SQLite database consulted manually by supervisors

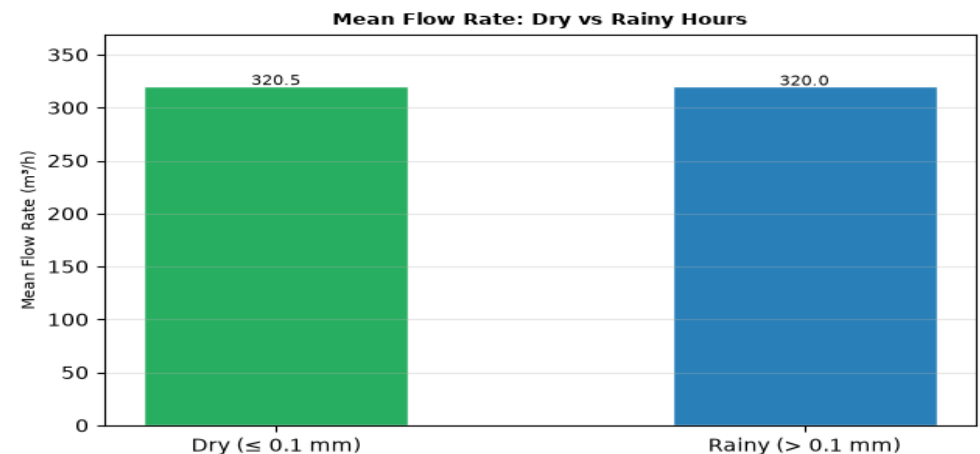
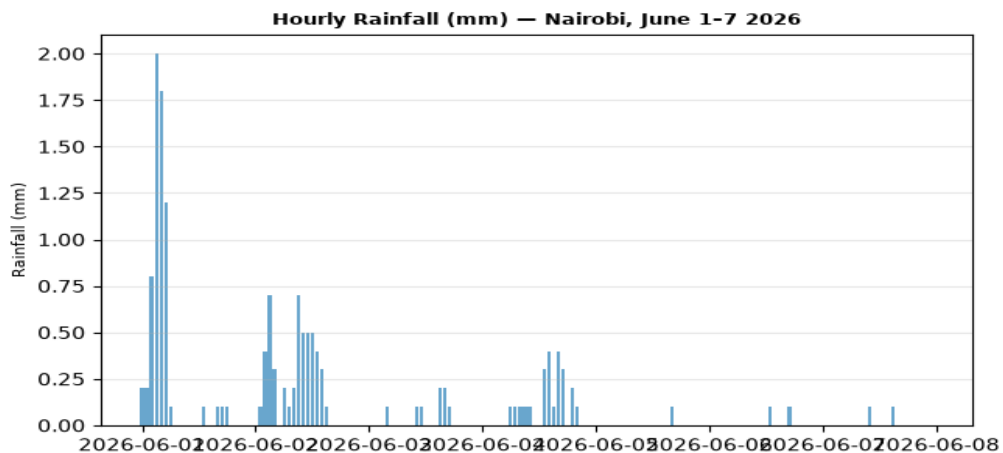
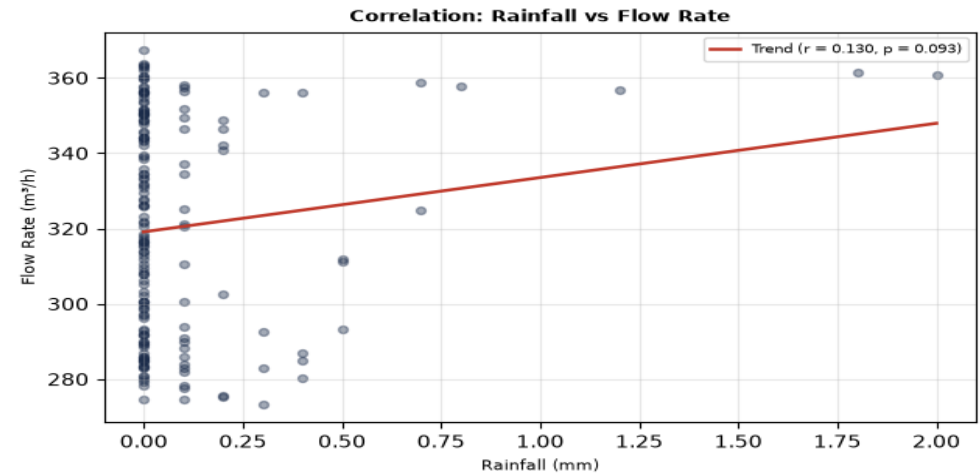
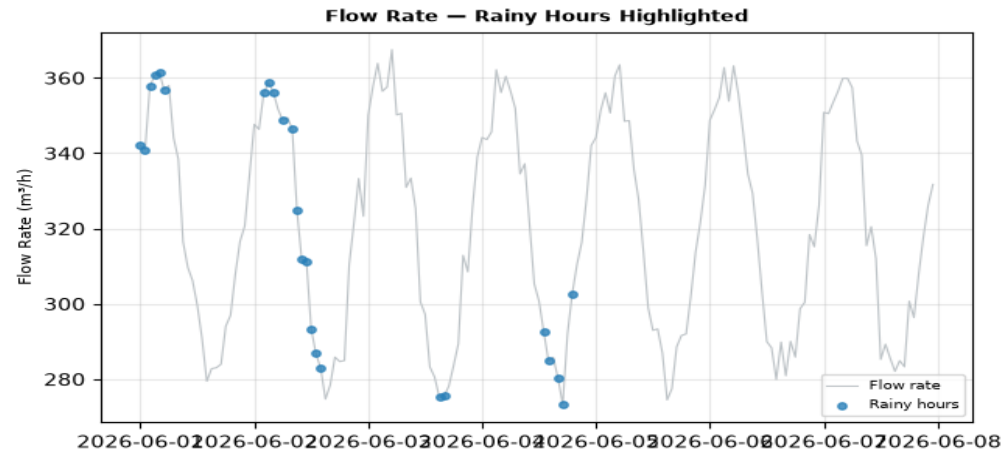
No one can ask: *"Did weather affect throughput? Was the June 1st dip a holiday effect or a real fault?"* — because the data is never in the same place at the same time.

The Pipeline Built

SOURCE 1	mock_ops.csv 1,002 rows cleaned 168 hourly after resample
SOURCE 2	Open-Meteo API 168 hourly rows precip · temp · wind
SOURCE 3	holiday_calendar.db 7 daily rows SQLite via <code>pd.read_sql()</code>
MASTER DF	Left join on timestamp 168 rows × 10 cols 0 nulls after merge

What the Merged Data Reveals

Week 3 — Multi-Source Analysis: Weather, Operations & Calendar



Panel A (top-left)

Flow rate time series with rainy hours as blue dots. Rainy hours cluster at the margins of daily cycles — not at the troughs.

Panel B (top-right)

Scatter: rainfall vs flow rate. Trend line $r = 0.130$, $p = 0.093$. Slope is positive and weak — rainfall does not suppress throughput.

Panels C & D (bottom)

Rainfall was light and intermittent (max 2.0 mm/hr). Dry hours averaged 320.5 m³/h vs rainy 320.0 m³/h — a 0.5 m³/h difference.

Findings & Operational Recommendations

CORRELATION (r) 0.130 Rainfall vs Flow Rate · $p = 0.093$	DRY vs RAINY (m³/h) 320.5 vs 320.0 0.5 m³/h difference — operationally negligible	MASTER DATAFRAME 168 rows 3 sources · 10 columns · 0 nulls
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What the Data Found

- Rainfall in Nairobi during the week of June 1–7 2026 was **light and intermittent** (max 2.0 mm/hr, mean 0.12 mm/hr)
- The Pearson correlation between rainfall and flow rate was **$r = 0.130$, $p = 0.093$** — weak, positive, and not statistically significant at the 95% confidence level
- **The hypothesis is not supported:** rainfall does not measurably suppress throughput over this 7-day period
- June 1 (Madaraka Day) and June 4 (Maintenance Shutdown) show **no significant flow rate difference** vs working days (320.2 vs 321.0 m³/h)
- The **nightly pressure spike** identified in Week 2 persists — and is not explained by weather or calendar effects

What to Do Next

- **Extend the observation window.** Seven days is insufficient to draw conclusions about rainfall effects. Re-run this pipeline over a full long-rains season (March–May) for a statistically powered test
- **Investigate the nightly pressure spike.** Weather and calendar have been ruled out as causes. The investigation should now focus on night-shift process changes and equipment scheduling
- **Automate this pipeline.** Week 4 will replace this manual notebook with a scheduled ETL that refreshes the master DataFrame daily without human intervention